CprE/SE 4910 WEEKLY REPORT 01

08/25/2026 – 09/22/2026

Group number: sdmay27-14

Project title: 3D Plan Visualization: Automated Building Reconstruction and Walkthrough Generator

Client: Buildertrend

Faculty Advisor: Dr. Islam

Team Members/Role:

- ***Nakshatra Gupta - Communication Lead***
- ***Rishiraj Parmar - Organization Lead***
- ***Ryan McNab - Team Meetings Lead***
- ***Daniel Hargrave - Computer Vision Lead***
- ***Jase Beaubien - Graphics Lead***

Weekly Summary

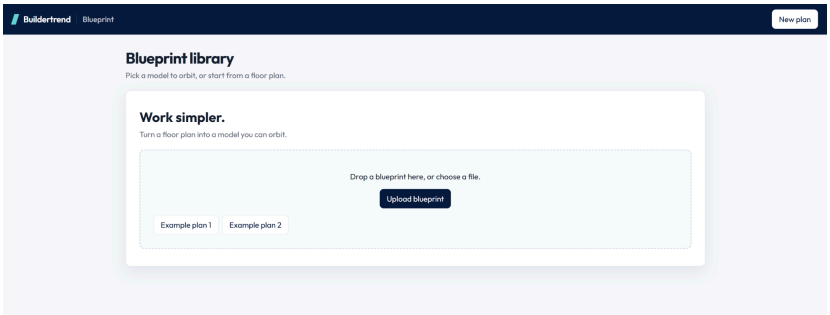
Most of the work during this timeframe was done as a group to ensure the whole team was on board. The overall objective was to become familiar with the project, decide team policies, delegate team roles, and discuss a general idea layout of the project. The roles decided are listed above. Additionally, after some discussion, it was decided to have two main groups for the project: the Computer Vision Team and the Graphics Team.

Past Week Accomplishments

- Group Work:
    - As discussed above, the team decided to have two main groups: the Computer Vision Team and the Graphics Team.
        - The Computer Vision Team will be tasked with designing a solution that takes in floor plans and elevation drawings in the form of various formats and converts them into information that the Graphics Team will be able to use.
        - The Graphics Team will be tasked with designing a solution that uses the information that was created by the Computer Vision Team and generates a 3D visualization with it. Additionally, hopefully the Graphics Team will also be able to design a viewer for the 3D model.

- Nakshatra Gupta:
    - Began researching how to structure the handoff between the Computer Vision Team and Graphics Team, since the project has been split into these two groups and the CV output needs to be directly usable by the Graphics pipeline.
    - Looked into common data interchange formats (JSON, CSV, and glTF for downstream 3D model export) to understand which would best represent parsed building data (walls, room boundaries, dimensions, etc.) in a way both teams could build against independently.

      http://linkinghub.elsevier.com/retrieve/pii/S0926580515002484
    - Reached out to the client and faculty advisor on behalf of the team. Sent an introductory email to the client contact outlining the team, expressing interest in the project, and requesting a meeting along with a preferred contact method and availability. Also followed up directly with the team's proposed faculty advisor to introduce the team.
- Rishiraj Parmar:
    - Researched methods for converting 2D PDF blueprints into 3D building models and explored how floor-plan information can be extracted and used for 3D reconstruction. Some articles that were reviewed:
        - Automatic Reconstruction of 3D Building Models from Scanned 2D Floor Plans
        - 3D reconstruction of building models based on 2D floorplans using statistical analysis
    - Began developing an understanding of the overall 2D-plan-to-3D-model pipeline and the information that will need to be transferred between the plan-processing and visualization components.
- Ryan McNab:
    - Started making a prototype off of the information provided.
        - Take a 2D image from a file picker.



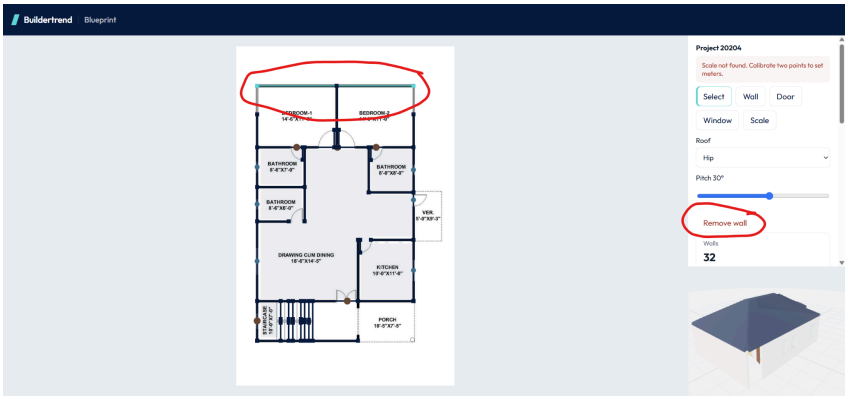
        - Clean up the image into black and white and a specified size, then show it to the user.



- Internally find exterior and interior walls as well as dimensions of said walls via the numbers on the image.
- Internally snap walls locations to be connected if close enough.
- Internally use gaps in walls next to curved lines to find locations of doors and windows.
- Internally find stairs by looking for many small rectangles near each other.
- Internally set base height of the floor to 2.7 meters and slop roof from exterior walls in multiple styles (Hip, Gable, Flat) with a variable slope.
- Turn each wall/door/roofing/stairs into a mesh composed of boxes, and then render a 3D visualization of said mesh into a small orbital viewer for the user to edit.



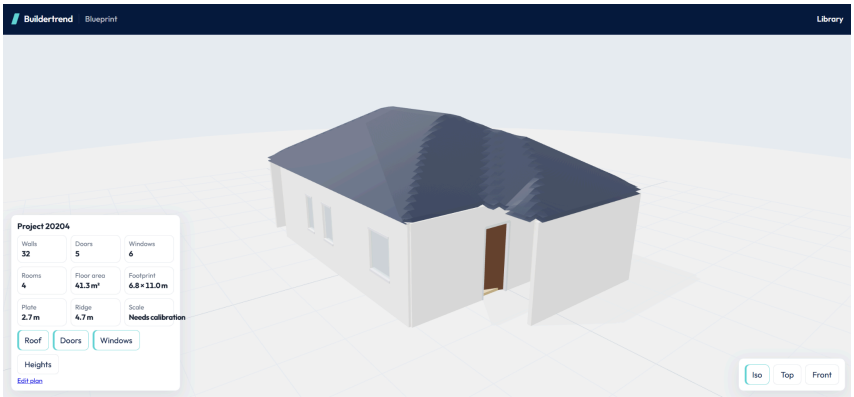
- The user selects walls/doors/windows to be able to delete/move/edit them.



- The user confirms the model looks good and saves the model to the library and it opens a larger orbital viewer of the model.



- Finally, the user drags and zooms around the model to get a good visual idea of the 2D plan.



- Daniel Hargrave:
  - Started researching computer vision with blueprints/floor plans. Read through [Exploring the Utility and Challenges of AI Interpretation of Construction Blueprints]. This helped to understand a lot more about the project. This includes some complications and details that will need to be known before getting started. Additionally, it provides a list of other research papers on the topic while describing what each of them is about. This will be extremely helpful going

forward with research for this project.
        - After reading this, it becomes extremely clear how important it is to know how the floor plans are made and what the plans look like.
        - The current rough idea is to make each page of the floor plans into two. The first will include all of the diagrams and symbols while the second will include any text that appears.
    - With the current information, the most likely language the computer vision section will be written in is Python and PyTorch will be used. The output will likely be either json or csv files. This could be different based on later requests from the Graphics Team.
- Jase Beaubien:
    - Began researching which graphics API would be ideal for the project's use-case. The team has already concluded the ideal choice is to build an in-house engine instead of using something like Unity or Unreal, but still have to decide between using OpenGL, Vulkan, or others as the graphics API.

Pending issues

- Nakshatra Gupta:
    - The exact interchange format cannot be finalized until the Computer Vision Team has a clearer picture of what the output will look like. Coordination between the two groups will be needed early to avoid designing the interface in isolation.
- Rishiraj Parmar:
    - More information is needed about the format and structure of the sample building plans before the interface between the computer vision and graphics components can be fully defined.
- Ryan McNab:
    - There are a lot of things to flesh out in terms of some portions of walls not rendering and furniture rendering as a bunch of walls instead of ignoring it. Also, the user interface could use a big upgrade, and there needs to be a set of example building plans to enhance further without wasting time.
- Daniel Hargrave:
    - A set of sample building plans are really needed. Knowing what types of plans that exist along with different symbols, text, and diagrams will be key in deciding how the project continues on the computer vision side.
- Jase Beaubien: N/A

Individual Contributions

NAME	**Individual Contributions** *(Quick list of contributions. This should be short.)*	**Hours this week**	**HOURS cumulative**
Nakshatra Gupta	Contributed to the team policy/leadership delegation for the team. Researched data interchange formats and began defining the shared data model connecting the Computer Vision Team's output to the Graphics Team's input.	4	4
Rishiraj Parmar	Contributed to the team policy/leadership delegation for the team. Researched methods for converting 2D PDF blueprints into 3D building models.	4	4
Ryan McNab	Contributed to the team policy/leadership delegation for the team. Also created an initial prototype to give the team a better visualization of the requirements.	5	5
Daniel Hargrave	Contributed to the team policy/leadership delegation for the team. Researched a decent about previous attempts of applying computer vision to blueprints/floor plans.	4	4
Jase Beaubien	Contributed to the team policy/leadership delegation for the team. Helped research graphics API options.	4	4

Comments and Extended Discussion

For a good chunk of the timeframe, the group worked on deciding the dynamics of the team and some of the initial requirements for getting started. For example, the group discussed and agreed upon having team meetings take place following the weekly Tuesday class time. Other things discussed were methods of communication, team policy, and leadership roles.

Additionally, the team discussed the purpose of the project and who is facing the problem that the project is trying to address. One of the decisions that was made following the discussion is that the group needs to put an emphasis on ease of use for the project. The solution should be able to be used by those who are not as skilled with blueprints or software. After discussing the types of people that the project is being made for, the group moved on to creating a rough roadmap for the project and the roles that each member should take on.

Plans for the upcoming week

- Nakshatra Gupta:
    - Meet with the team to understand expected inputs and outputs, then draft an initial shared data schema that both groups can review. Research glTF and other 3D-friendly formats to determine the best target for the Graphics Team's input.
- Rishiraj Parmar:
    - Research methods for converting extracted 2D floor-plan geometry into structured data that can later be used for 3D model generation.
    - Investigate possible intermediate data formats, such as JSON, for transferring building geometry between the Computer Vision and Graphics components.
- Ryan McNab:
    - Improving the prototype to fix as many bugs as possible and make it look more professional.
    - Try meeting with our advisor and Buildertrend.
- Daniel Hargrave:
    - Research more about use of computer vision with regards to blueprints/floor plans.
    - See if the idea to extract the text from the blueprint and make it its own separate page is feasible and would help.
    - Potentially start working on writing python code that will be used for the computer vision group.
- Jase Beaubien:
    - Solidify graphics API choice.
    - Get a foundation set in the repository (non-invasive architecture, i.e. just opening a window and/or rendering a test mesh).